GLOBAL POLLUTION ANALYSIS AND ENERGY RECOVERY

Apriori Association Rule Mining

Dataset: Global_Pollution_Analysis.csv

1. Introduction

This project studies pollution levels and energy-related information using association rule mining. The main purpose is to find combinations of pollution severity and energy conditions that occur frequently in the dataset. Apriori is used to generate frequent itemsets and association rules.

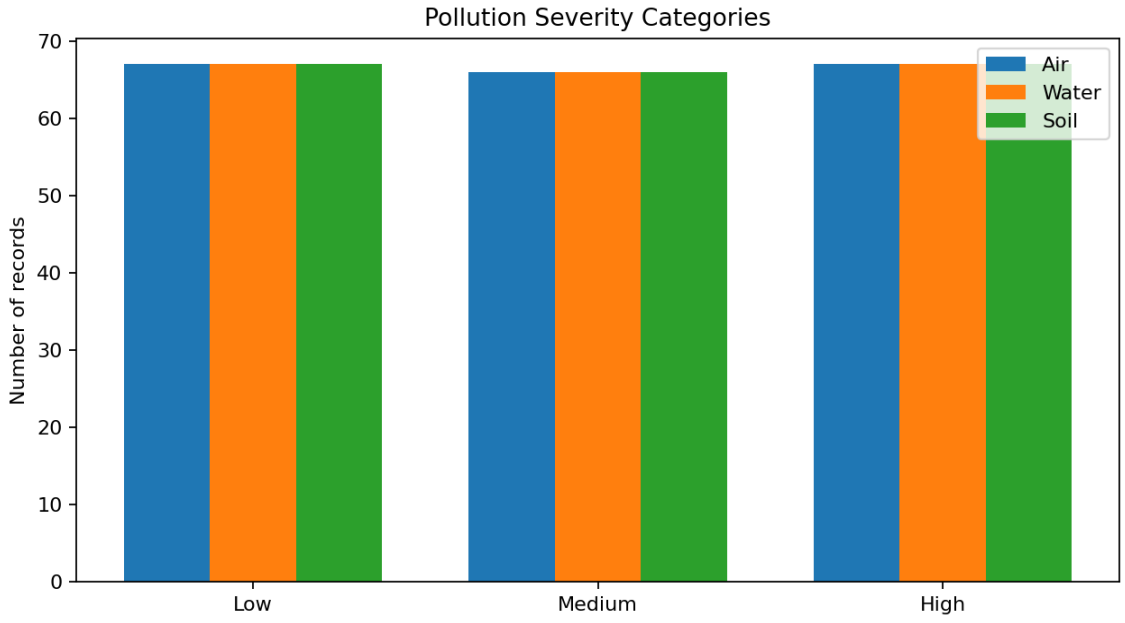
2. Dataset Loading and Preprocessing

The dataset contains 200 records and 13 columns. It includes air, water and soil pollution indexes, industrial waste, energy recovered, CO2 emissions, renewable energy, plastic waste, energy consumption per capita, population and GDP per capita. The data was loaded using Pandas, duplicate rows were removed and numerical columns were checked. After preprocessing, there were no missing values in the working data.

Item	Value
Rows after preprocessing	200
Columns	13
Countries	175
Years	2000–2019
Missing values	0
Duplicate rows	0

3. Pollution Severity Categorization

The air, water and soil pollution indexes were divided into Low, Medium and High categories. The 33rd and 67th percentiles were used as the boundaries. This makes the numerical pollution values suitable for transaction-based association rule mining.



4. Energy Per Capita Feature Engineering

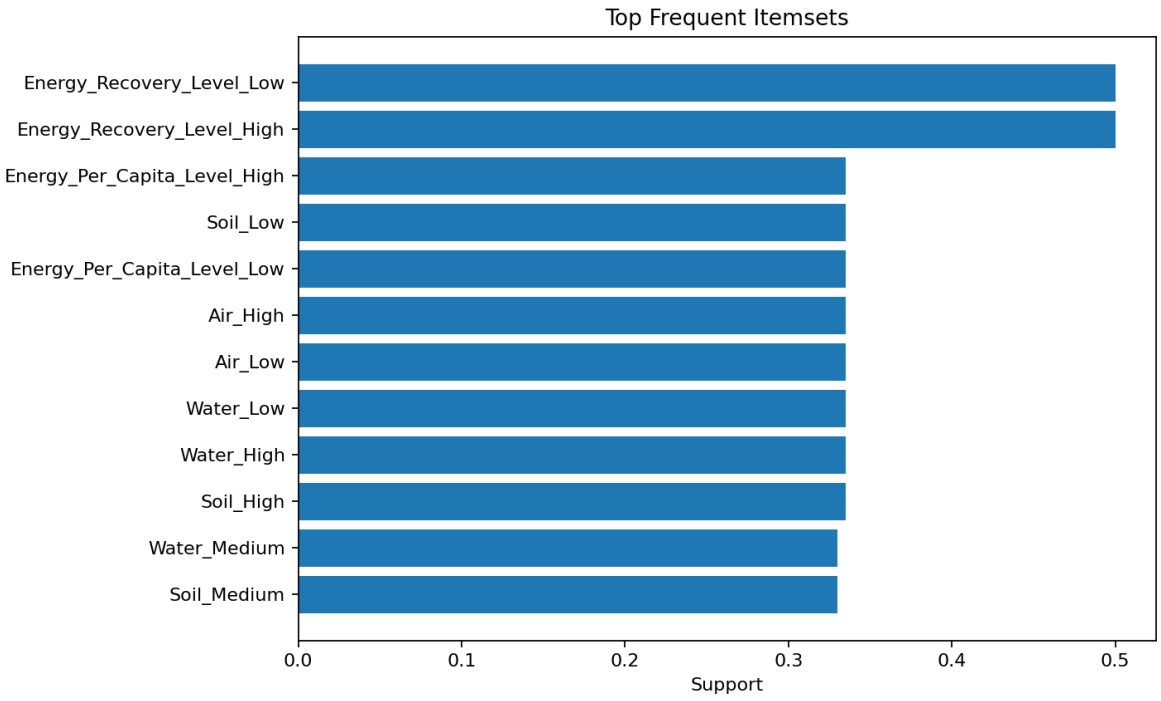
The dataset already contains `Energy_Consumption_Per_Capita (in MWh)`, so it was retained as the energy-per-capita feature. For Apriori, it was also converted into Low, Medium and High categories using the 33rd and 67th percentiles. Energy recovered was divided into High and Low using its median value.

5. Transaction Creation

Each row was treated as one transaction. Items include pollution severity and energy categories, for example Air_High, Water_Medium, Soil_Low and Energy_Per_Capita_High. This format allows the Apriori algorithm to search for combinations that occur repeatedly.

6. Apriori Algorithm Implementation

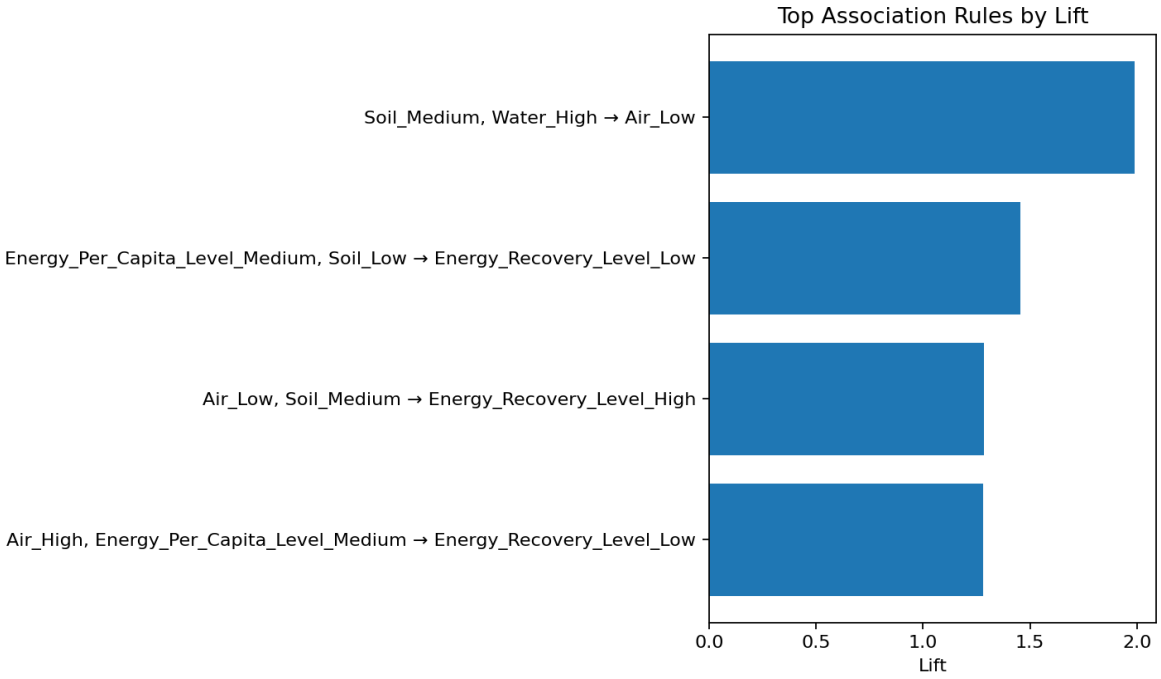
A direct Python implementation of Apriori was used with minimum support = 0.08. The algorithm finds frequent single items, joins them to form larger candidates, applies Apriori pruning, and keeps itemsets meeting the support threshold. 96 frequent itemsets were found.



7. Association Rule Mining

Rules were generated using minimum confidence = 0.60. For each rule, support, confidence and lift were calculated. Rules below the confidence threshold were removed.

Antecedent	Consequent	Support	Confidence	Lift
Soil_Medium, Water_High	Air_Low	0.080	0.667	1.990
Energy_Per_Capita_Level_Medium, Soil_Low	Energy_Recovery_Level_Low	0.080	0.727	1.455
Air_Low, Soil_Medium	Energy_Recovery_Level_High	0.090	0.643	1.286
Air_High, Energy_Per_Capita_Level_Medium	Energy_Recovery_Level_Low	0.080	0.640	1.280



8. Evaluation Using Support, Confidence and Lift

Support measures how often the complete item combination appears. Confidence measures how often the consequent appears when the antecedent appears. Lift compares the observed confidence with the confidence expected from the individual occurrence of the consequent. A lift above 1 indicates a positive association in the transaction data.

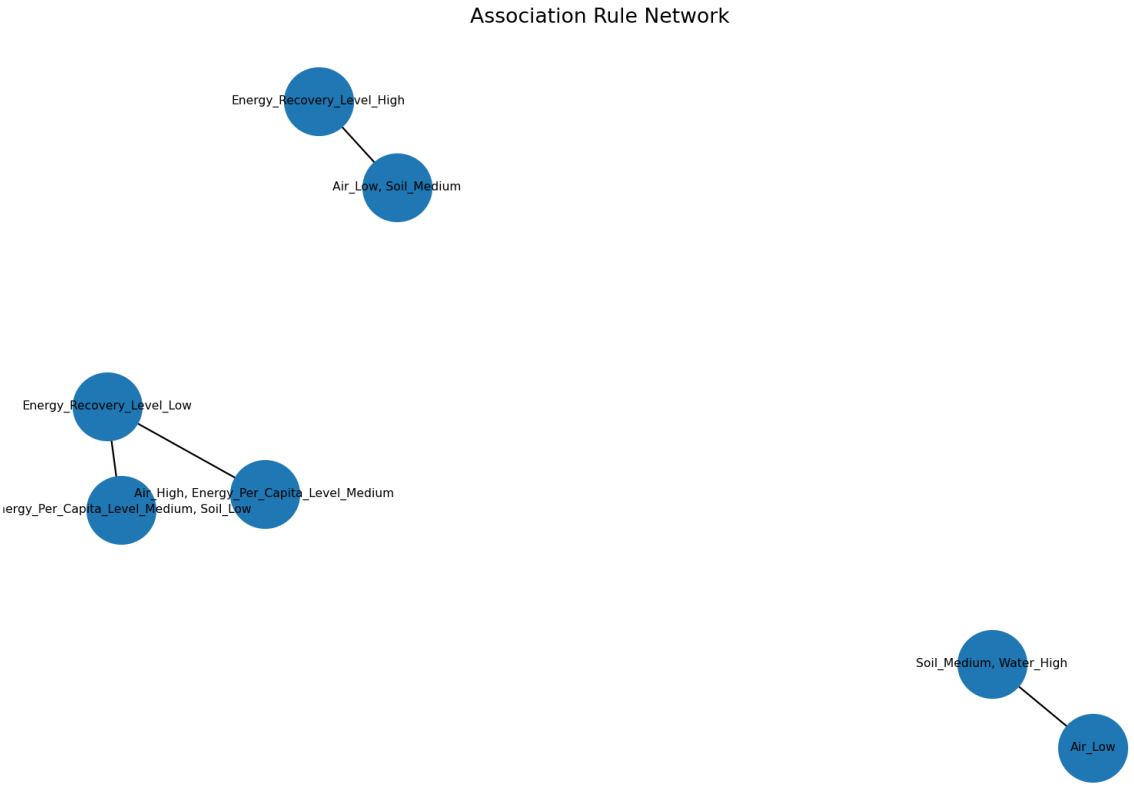
9. Rule Cross-Validation

The transactions were divided into five folds. The strongest discovered rules were checked on each held-out fold. Mean support, mean confidence and mean lift were calculated across the folds. This gives a simple stability check for the association rules.

Rule	Mean Support	Mean Confidence	Mean Lift	Valid Folds
Soil_Medium, Water_High → Air_Low	0.080	0.674	2.007	5
Energy_Per_Capita_Level_Medium, Soil_Low → Energy_Recovery_Level_Low	0.080	0.787	1.575	5
Air_High, Energy_Per_Capita_Level_Medium → Energy_Recovery_Level_Low	0.080	0.686	1.390	5
Air_Low, Soil_Medium → Energy_Recovery_Level_High	0.090	0.659	1.287	5

10. Association Rule Network

The network shows the strongest rules as directed connections. The arrow points from the antecedent to the consequent.



11. Conclusion

The project covers all the main steps required for the pollution and energy analysis. The dataset was loaded and cleaned, pollution severity was categorized, energy per capita was included as a feature, and Apriori was implemented to generate frequent itemsets. Association rules were filtered using support and confidence and evaluated using lift. Five-fold rule validation and visualizations were also included. The discovered rules show associations in the dataset and should not be interpreted as proof of cause and effect.

12. Software Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and NetworkX.